Timing-Content Separation for Human-Coach-Like Exercise Feedback Generation

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Abstract

Prior work has shown that 3D pose and biomechanical features can support language feedback for exercise coaching. The AI Coach Challenge: Fitness evaluates this setting through sparse, short, actionable cues that must match ground-truth coaching feedback in both timing and content. In our preliminary experiments, a single end-to-end video-to-feedback baseline did not jointly improve feedback timing and content quality, suggesting that deciding when to speak and what to say should be handled separately. We propose FitCoachPipe, a timing/content pipeline in which a speak gate with a 3-second cooldown selects speaking moments, and a supervised-finetuned Qwen3.5 content model generates feedback at the selected timestamps from pose-derived observations, utterance history, and visual context. FitCoachPipe achieves a T-F of 0.586 and an LLM-Acc of 3.093 on our local benchmark, and a T-F of 0.727 and an LLM-Acc of 2.809 on the official competition evaluation. We further analyze sparse body-aware feedback, cooldown-based over-emission control, and the roles of history and longer visual context.

1. Introduction

Video-based exercise feedback has the potential to support sports training, rehabilitation, and everyday fitness [1, 2, 5, 6], yet effective coaching depends on giving the right cue at the right moment. The AI Coach Challenge: Fitness [7] captures this setting by evaluating feedback utterances through both timestamp and content matching to ground-truth coaching feedback. Content fluency alone is therefore insufficient, since correct advice may be penalized if it is given at the wrong time, while well-timed but generic advice may fail to match the intended coaching content.

Prior work on fitness feedback and form assessment has shown that pose, 3D motion, and biomechanical features can support interpretable exercise analysis and language feedback [1–3, 5]. We also use detailed pose information, but treat it as internal evidence for deciding what to say and whether to speak, rather than as text to verbalize directly. In our preliminary experiments, a single end-to-end video-to-feedback baseline did not jointly improve feedback timing and content quality, suggesting that these two decisions should be handled separately.

We introduce FitCoachPipe, a timing/content pipeline that separates when to speak from what feedback to provide. In the timing stage, a speak gate with a 3-second cooldown selects speaking moments, suppressing nearby duplicate outputs while preserving plausible feedback opportunities. In the content stage, FitCoachPipe generates short feedback at the selected timestamps using pose-derived observations, utterance history, and skeleton-based form-error cues.

On `localBenchmark`, FitCoachPipe improves over the video-only Qwen3.5 baseline in both timing and content, raising T-F from 0.495 to 0.586 and LLM-Acc from 2.960 to 3.093. Compared with the timing-only gate, it preserves the same T-F of 0.586 while improving LLM-Acc from 2.768 to 3.093. On the official competition evaluation, FitCoachPipe obtains METEOR 0.167, ROUGE-L 0.122, BERT-F1 0.888, LLM-Acc 2.809, and T-F 0.727. We further analyze the effects of body-aware feedback, cooldown-based over-emission control, history, and longer visual context.

2. Method

FitCoachPipe separates exercise coaching into pose-based observation, timing selection, and feedback generation. Given an input video, the system first extracts body-aware observations from 3D pose, then selects when to speak, and finally generates short advice only at the selected timestamps. Figure 1 summarizes this timing-content separation. This separation avoids entangling utterance count, timestamp alignment, and content specificity in a single model.

Body-aware Observations. We estimate 3D body pose using Sam3DPose (SAM 3D Body) [8] and convert it into textual and symbolic observations. These observations describe posture, exercise state, movement phase, and likely form errors, and are shared by the timing and content stages. We use them as internal evidence for coaching decisions

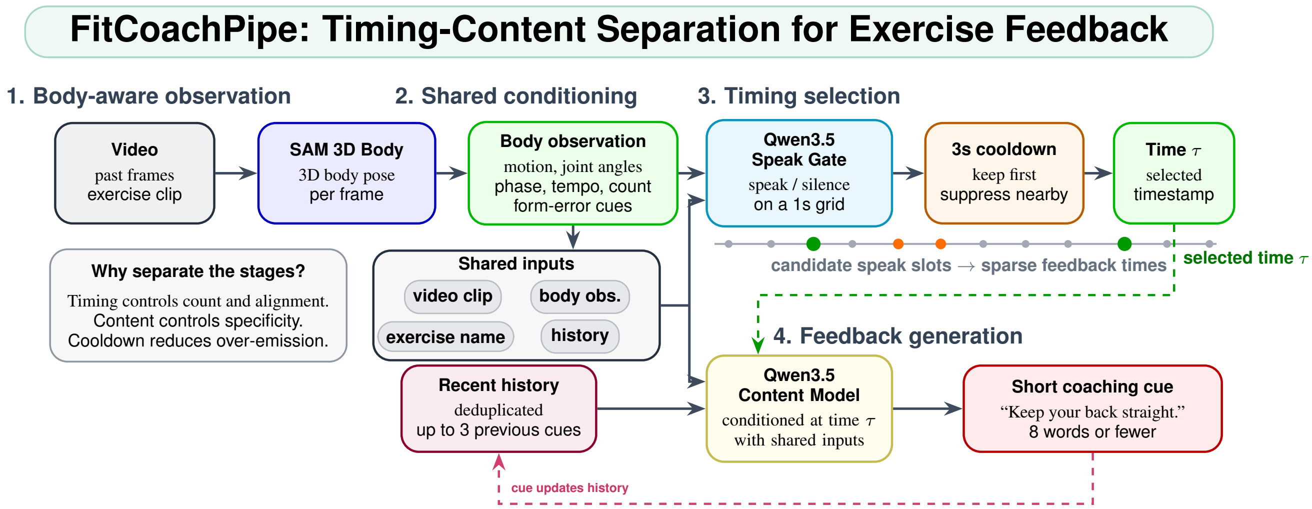

Figure 1. **FitCoachPipe overview.** The pipeline separates when to speak from what to say: body-aware observations feed a timing gate and cooldown filter to select sparse feedback times, while shared inputs and recent history condition a short coaching cue at the selected timestamp.

rather than directly verbalizing pose measurements or joint-angle descriptions. At each 1-second video slot, we construct a causal pose-derived observation from frames ending at that slot. It contains 3-second motion energy and trend, recent joint-angle statistics, rule-based severity and persistence labels, and repetition phase, tempo, and count cues.

Timing Selection. The timing gate is a supervised Qwen3.5 SFT model that receives the causal video and the pose-derived observation on a 1-second grid and predicts whether the coach should speak at each slot. For training, we label slots whose centers fall within 1.5 seconds of each ground-truth feedback timestamp as positive, rather than requiring an exact single-slot match, so the gate learns recall-oriented coaching opportunities. At inference time, the emitted speak decisions are treated as candidate timestamps, and a 3-second cooldown filter keeps the first candidate while suppressing subsequent candidates within the cooldown window.

Feedback Generation. At each selected timestamp, a supervised-finetuned Qwen3.5 vision-language model generates one short coaching cue. It receives the same pose-derived body-aware observation used by the timing gate, recomputed at that timestamp, along with the exercise name, a video clip ending at the decision time, and up to three recent deduplicated utterances. We constrain feedback to eight words or fewer, use a 24-second maximum visual context window, and update the history with the model's own outputs.

3. Experiments

3.1. Setup

Evaluation Metrics. We evaluate our system following the QEVD-FIT-COACH Benchmark [4]. The primary score is T-F, which measures whether predicted feedback matches the ground-truth coaching feedback in both timing and text similarity. We also report METEOR, ROUGE-L, BERT-F1, and 1–5 LLM-Acc, where higher values indicate better performance.

Implementation Details. Our main configuration uses FitCoachPipe with the 3-second cooldown timing schedule, a 24-second past-only visual context, up to three deduplicated previous utterances as history, and the supervised-finetuned Qwen3.5 content model. We compare this setting with variants that remove utterance history, shorten the visual context to 5 or 16 seconds, or combine history removal with the 5-second context. These comparisons isolate the effects of feedback history and longer visual context on timing-aware exercise coaching.

3.2. Results

Local benchmark comparison. Table 1 shows the main comparison on `localBenchmark`. The video-only Qwen3.5 SFT baseline achieves T-F 0.495 and LLM-Acc 2.960, while adding pose-derived observations improves both scores to T-F 0.549 and LLM-Acc 3.066. The timing gate further raises T-F to 0.586, but its lower LLM-Acc of 2.768 indicates that timing alone is insufficient for high-quality coaching feedback. FitCoachPipe keeps this strong timing score and improves content quality, achieving

Method	Mode	METEOR	ROUGE-L	BERT-F1	LLM-Acc	T-F
Qwen3.5 video-only	V	0.154	0.143	0.884	2.960	0.495
Qwen3.5 (intv.=3s)	V+P	0.171	**0.160**	0.887	3.066	0.549
BioCoach	V+B	0.129	0.122	0.864	2.560	0.544
Timing-only gate	V+P+G+CD	0.119	0.109	0.882	2.768	**0.586**
FitCoachPipe	**V+P+C**	**0.187**	0.152	**0.890**	**3.093**	**0.586**

Table 1. Main results on `localBenchmark`. Mode codes: V = video, P = pose-derived observations, B = biomechanics, G = speak gate, CD = cooldown, C = fitness-base content model.

Setting	METEOR	ROUGE-L	BERT-F1	LLM-Acc	T-F
Input ablation with fixed-grid Qwen					
Video only	0.154	0.143	0.884	2.960	**0.495**
Video + pose observations	**0.174**	**0.162**	**0.888**	**3.008**	0.493
History ablation with 24s context					
With history	0.1866	0.1518	**0.8898**	**3.0927**	**0.5862**
Without history	**0.1974**	**0.1539**	0.8888	3.0845	0.5851
Context-length ablation with history					
5s context	0.1679	0.1260	0.8865	**3.0965**	0.5857
16s context	0.1793	0.1399	0.8886	3.0407	0.5861
24s context	**0.1866**	**0.1518**	**0.8898**	3.0927	**0.5862**
Combined ablation					
Without history + 5s context	0.1659	0.1313	0.8829	2.9192	0.5325

Table 2. Ablations on pose input, utterance history, and visual context length.

the best METEOR 0.187, BERT-F1 0.890, and LLM-Acc 3.093, with a tied-best T-F of 0.586. BioCoach is shown as a reference from prior work, since we did not rerun it under our implementation.

Official submission result. FitCoachPipe obtains METEOR 0.167, ROUGE-L 0.122, BERT-F1 0.888, LLM-Acc 2.809, and T-F 0.727. This result shows that the submitted timing/content pipeline achieves a high official T-F score while maintaining reasonable content quality under the competition evaluation.

Ablations. Table 2 summarizes ablations on pose input, utterance history, and visual context length. In the fixed-grid Qwen comparison, adding pose-derived observations improves the content metrics, increasing METEOR from 0.154 to 0.174, ROUGE-L from 0.143 to 0.162, BERT-F1 from 0.884 to 0.888, and LLM-Acc from 2.960 to 3.008, while T-F remains nearly unchanged. For FitCoachPipe with a 24-second context, removing history improves METEOR and ROUGE-L but slightly reduces BERT-F1, LLM-Acc, and T-F, suggesting a trade-off between lexical overlap and semantic consistency. Increasing the context length from 5s to 24s steadily improves METEOR, ROUGE-L, and BERT-F1, while T-F remains stable around 0.586. The combined setting without history and with only 5s context drops to LLM-Acc 2.919 and T-F 0.533, indicating that short context and missing history together hurt semantic feedback quality and timestamp-content matching.

Qualitative Diagnostics. Qualitative diagnostics further clarify the aggregate scores. Fig. 2 compares the Qwen3.5 SFT video-only baseline, FitCoachPipe, and two FitCoachPipe ablations on the same floor-touch episode. The Qwen3.5 SFT video-only baseline tends to produce generic count feedback after posture changes, whereas FitCoachPipe, our proposed method, adapts its feedback from reach-lower correction to progress and count-related feedback. The history-off row remains locally plausible at matched timestamps, consistent with its competitive lexical scores, but gives less evidence of episode-level progression. The 5s context row shows a related failure mode: it produces plausible local posture cues early, but later drifts into premature repetition counts, supporting the aggregate finding that longer context improves text-quality metrics.

4. Discussion

The results show that timing and content are difficult to optimize in one output sequence. Qwen3.5 SFT interval=3.0s has strong text metrics and LLM-Acc but lower T-F than FitCoachPipe, while the timing-only gate improves timing but has limited content quality. Separating timing from content, with pose-derived observations, cooldown, history, and longer visual context, is therefore a practical design for sparse body-grounded coaching feedback.

The ablations support this view. Pose-derived observations improve text metrics and LLM-Acc in the historical comparison, history trades surface overlap for contextual consistency, and longer visual context improves n-gram metrics. The clearest degradation appears only when both history and long context are removed, suggesting complementary information paths.

5. Conclusion

We frame AI Coach Challenge: Fitness as producing short, human-coach-like utterances at appropriate moments, using pose as internal evidence for timing, error selection, and concise wording. Because one model did not improve timing and content together, FitCoachPipe connects a cooldown-based timing stage with feedback generation. FitCoachPipe achieves T-F 0.586 and LLM-Acc 3.093 on localBenchmark, and T-F 0.727 and LLM-Acc 2.809 on the official competition score, supporting concise feedback grounded in body-state evidence.

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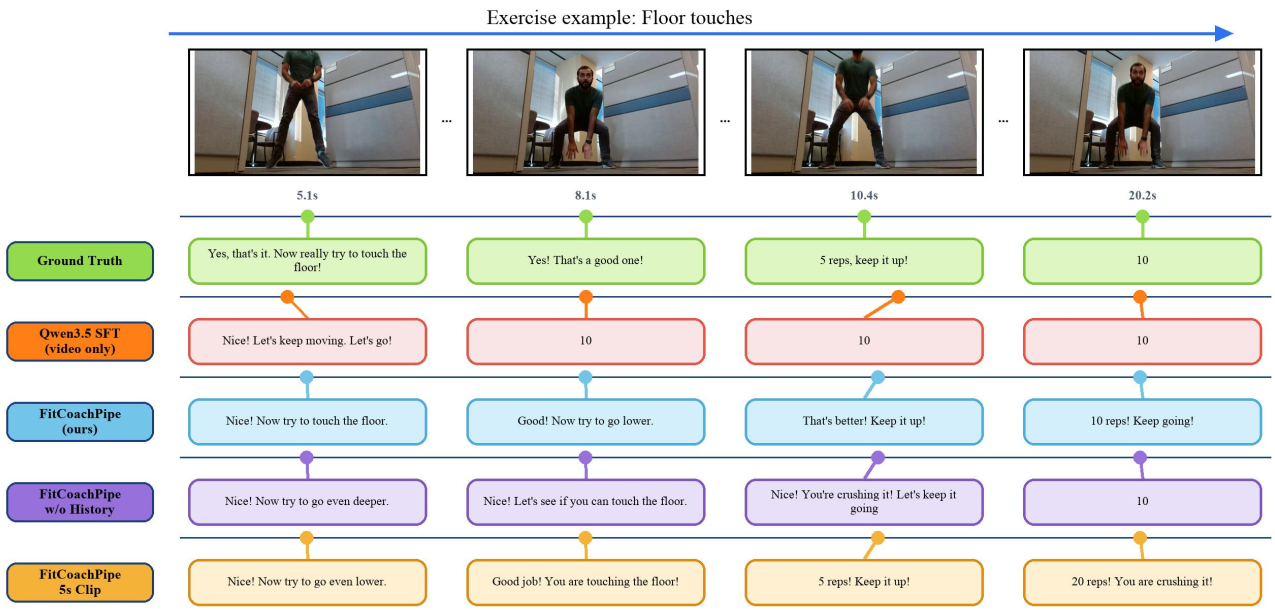

Figure 2. Qualitative floor-touch ablation. We compare ground truth, the Qwen3.5 SFT video-only baseline, FitCoachPipe, and two FitCoachPipe ablations without utterance history or with only 5s visual context at matched timestamps.

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